Introducing the concepts of a metric and a space

Introducing total variation distance as a distance between distributions (9.1)

The concept of coupling (9.2)

Its relation to d_{TV}.

Introducing Wasserstein distance (12.4)

Calculating Wasserstein distance using the quantile function.

To understand the concepts in this section better, we first clarify the two notions of a "space" and a "metric."

Space: An informal definition

A space is a set on which a specific structure is defined, and in which relations among the members of the set are defined.

Metric

If we consider a set A, a function of the form

$$d : A \times A \to [0, \infty)$$

is called a metric on A if, for every $x, y, z \in A$, it has the following three properties, the properties of distance:

i)	$x = y \iff d(x, y) = 0$	(positive definiteness),
ii)	$d(x, y) = d(y, x)$	(symmetry),
iii)	$d(y, z) + d(x, y) \geq d(x, z)$	(triangle inequality).

Section 9.1: Total variation distance

The **total variation distance** between probability measures μ_1, μ_2 on a measurable space $(S, \mathcal{S})$ is defined by

$$d_{\mathrm{TV}}(\mu_1, \mu_2) = \sup_{A \in \mathcal{S}} |\mu_1(A) - \mu_2(A)|.$$

Intuitively, for all possible events, we examine the difference between their probabilities under μ_1 and μ_2. We then take the greatest difference as d_{TV}.

Proposition. If $P(S, \mathcal{S})$ is the set of probability measures on $(S, \mathcal{S})$, then total variation distance d_{TV} is a metric on $P(S, \mathcal{S})$.

Intuitively, we can regard the probability measures on $(S, \mathcal{S})$ as points in a geometric space. The definition of d_{TV} is a distance between these points.

Proof. From the definition of d_{TV}, it is easy to show that it has all three properties of a metric.

Example. The value of d_{TV} between the Dirac measures δ_x, δ_y on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is

$$\begin{cases} x \neq y & \Rightarrow \quad d_{\mathrm{TV}} = 1, \\ x = y & \Rightarrow \quad d_{\mathrm{TV}} = 0. \end{cases}$$

In the first case, there exists $A \in \mathcal{B}(\mathbb{R})$ such that $x \in A$, $y \notin A$, so $\delta_x(A) = 1$ and $\delta_y(A) = 0$.

Lemma

For measurable functions $f_1, f_2 : S \to \mathbb{R}_+$ such that

$$\int_S f_2 \, d\nu = \int_S f_1 \, d\nu = 1,$$

we have

$$\text{i)} \quad \int_S (f_1 - f_2)_+ \, d\nu = \int_S (f_2 - f_1)_+ \, d\nu = \frac{1}{2} \int_S |f_1 - f_2| \, d\nu,$$
$$\text{ii)} \quad \int_S (f_1 \wedge f_2) \, d\nu = 1 - \frac{1}{2} \int_S |f_1 - f_2| \, d\nu,$$

where

$$f_1 \wedge f_2 := \min\{f_1, f_2\}, \qquad (f_1 - f_2)_+ := \max\{f_1 - f_2, 0\}.$$

Intuition. This lemma says that, if two densities f_1, f_2 are defined on a probability space, the excess mass of f_1 relative to f_2 is exactly half the total difference in mass, and their common mass equals one minus half that total difference.

Proof. i) Define $A := \{x : f_1(x) > f_2(x)\}$. Then, by definition,

$$(f_1 - f_2)_+ = (f_1 - f_2)\mathbf{1}_A, \qquad (f_2 - f_1)_+ = (f_2 - f_1)\mathbf{1}_{A^c}.$$

Taking integrals of both sides gives

$$\int_S (f_1 - f_2)_+ \, d\nu = \int_A (f_1 - f_2) \, d\nu, \tag{1}$$
$$\int_S (f_2 - f_1)_+ \, d\nu = \int_{A^c} (f_2 - f_1) \, d\nu. \tag{2}$$

Since $\int_S f_1 \, d\mu = \int_S f_2 \, d\mu$,

$$0 = \int_S (f_2 - f_1) \, d\mu = \int_A (f_2 - f_1) \, d\mu + \int_{A^c} (f_2 - f_1) \, d\mu.$$

Therefore,

$$\int_A (f_1 - f_2) \, d\mu = \int_{A^c} (f_2 - f_1) \, d\mu.$$

The right-hand sides of (1) and (2) are equal, so

$$\int_S (f_2 - f_1)_+ \, d\nu = \int_S (f_1 - f_2)_+ \, d\nu. \tag{*}$$

It is also clear that

$$|a - b| = (a - b)_+ + (b - a)_+$$

Thus,

$$\int_S |f_1 - f_2| \, d\nu = \int_S (f_2 - f_1)_+ \, d\nu + \int_S (f_1 - f_2)_+ \, d\nu.$$

Using (*),

$$\int_S (f_1 - f_2)_+ \, d\nu = \int_S (f_2 - f_1)_+ \, d\nu = \frac{1}{2} \int_S |f_1 - f_2| \, d\nu.$$

ii)

$$\begin{aligned}
\int_S (f_1 \wedge f_2) \, d\nu &= \int_A f_2 \, d\nu + \underbrace{\int_{A^c} f_1 \, d\nu}_{1 - \int_A f_1 \, d\nu} \\
&= 1 - \int_A (f_1 - f_2) \, d\nu \\
&= 1 - \int_S (f_1 - f_2)_+ \, d\nu \\
&= 1 - \frac{1}{2} \int_S |f_1 - f_2| \, d\nu,
\end{aligned}$$

where the last equality follows from the preceding part.

Proposition

If μ_1, μ_2 are the laws of densities $f_1, f_2 : S \to \mathbb{R}_+$, respectively, with respect to a reference measure ν on the probability space $(S, \mathcal{S})$, then

$$d_{\mathrm{TV}}(\mu_1, \mu_2) = \frac{1}{2} \int_S |f_1(x) - f_2(x)| \, \nu(dx).$$

In the particular case where S is countable,

$$d_{\mathrm{TV}}(\mu_1, \mu_2) = \frac{1}{2} \sum_{x \in S} |f(x_1) - f(x_2)|.$$

Intuition. The TVD between the two distributions μ_1, μ_2 is half the difference between their masses in the space.

Proof. We use the following idea:

$$a = b \quad \Rightarrow \quad a \geq b, \quad a \leq b.$$

Set $A := \{x : f_1(x) > f_2(x)\}$. Then

$$\begin{aligned}
\text{i)} \quad \int_S (f_1 - f_2)_+ \, d\nu &= \int_A f_1 \, d\nu - \int_A f_2 \, d\nu \\
&= \mu_1(A) - \mu_2(A) \\
&\leq |\mu_1(A) - \mu_2(A)| \leq d_{\mathrm{TV}}(\mu_1, \mu_2),
\end{aligned}$$

where the last inequality follows from the definition of d_{TV}.

Thus $\int_S (f_1 - f_2)_+ \, d\nu \leq d_{\mathrm{TV}}(\mu_1, \mu_2)$. By the preceding lemma,

$$\frac{1}{2} \int_S |f_1 - f_2| \, d\nu \leq d_{\mathrm{TV}}(\mu_1, \mu_2). \tag{1}$$

Also, for any set B,

$$(f_1 - f_2)\mathbf{1}_B \leq (f_1 - f_2)_+ \mathbf{1}_B \leq (f_1 - f_2)_+,$$

since $f_1 - f_2 \leq (f_1 - f_2)_+$.

Thus, for any set B, we have

$$\begin{aligned}\mu_1(B) - \mu_2(B) &= \int_B f_1 \, d\nu - \int_B f_2 \, d\nu \\ &= \int_S (f_1 - f_2)\mathbf{1}_B \, d\nu.\end{aligned}$$

Consequently,

$$\mu_1(B) - \mu_2(B) \leq \int_S (f_1 - f_2)_+ \, d\nu = \frac{1}{2}\int_S |f_1 - f_2| \, d\nu,$$

where the equality follows from the previous lemma. Similarly,

$$\mu_2(B) - \mu_1(B) \leq \int_S (f_2 - f_1)_+ \, d\nu = \frac{1}{2}\int_S |f_1 - f_2| \, d\nu.$$

Therefore,

$$|\mu_2(B) - \mu_1(B)| \leq \frac{1}{2}\int_S |f_1 - f_2| \, d\nu.$$

Since this holds for every $B \in \mathcal{S}$, we can conclude that

$$d_{\mathrm{TV}}(\mu_1, \mu_2) \leq \frac{1}{2}\int_S |f_1 - f_2| \, d\nu. \tag{2}$$

From (1) and (2),

$$d_{\mathrm{TV}}(\mu_1, \mu_2) = \frac{1}{2}\int_S |f_1 - f_2| \, d\nu.$$

To understand this better, we examine an example.

Example: TVD between two Bernoulli distributions

Determine the TVD between $\mathrm{Ber}(p)$ and $\mathrm{Ber}(q)$.

Solution.

$$\begin{aligned}d_{\mathrm{TV}} &= \frac{1}{2}\sum_{x \in S} |f_1(x) - f_2(x)| \\ &= \frac{1}{2}(|f_1(0) - f_2(0)| + |f_1(1) - f_2(1)|) \\ &= \frac{1}{2}(|(1-p) - (1-q)| + |p - q|) = \boxed{|p - q|}.\end{aligned}$$

An intuitive example

Suppose that the people in a community have been divided into three groups according to income, and that the population distributions of these groups have been measured at two different times, denoted by μ_1, μ_2.

$$\begin{array}{rccc} & \text{Low} & \text{Medium} & \text{High} \\ \mu_1 = & (0.2, & 0.6, & 0.2) \\ & \downarrow +0.3 & \downarrow -0.4 & \downarrow +0.1 \\ \mu_2 = & (0.5, & 0.2, & 0.3) \end{array}$$

Here we have two distributions on the same space, a community. Their distance is meaningful, so we can compare the two distributions:

$$\begin{aligned}d_{\mathrm{TV}} &= \frac{1}{2}(|0.5 - 0.2| + |0.2 - 0.6| + |0.3 - 0.2|) \\ &= 0.4.\end{aligned}$$

(Calculate d_{TV} from its general definition as well.)

Here, 0.4 units of mass have been moved. This is exactly the meaning of total variation distance.

Section 9.2: Coupling

A **coupling** of probability measures μ_1 on $(S_1, \mathcal{S}_1)$ and μ_2 on $(S_2, \mathcal{S}_2)$ is a pair (X_1, X_2) of random variables defined on a common probability space $(\Omega, \mathcal{A}, \mathbb{P})$ such that the distribution of X_1 is μ_1 and the distribution of X_2 is μ_2.

In brief, a coupling is a joint probability distribution of two random variables on the joint space $S_1 \times S_2$, whose marginal distributions are μ_1, μ_2. Their dependence is determined by the coupling.

Thus, the set of couplings represents the set of all possible dependencies between two probability measures. An independent coupling also represents independence.

Intuition. Suppose that we have a group of people and know their height and weight distributions. These are the marginal distributions. Now consider the joint space $(\text{weight}, \text{height})$. A coupling specifies how to choose a pair such as $(180, 65)$ so that its marginal distributions are still μ_1, μ_2. With this choice, we actually obtain information about dependence between height and weight.

Proposition

The total variation distance between probability measures μ_1, μ_2 on a measurable space $(S, \mathcal{S})$ is bounded as follows:

$$\boxed{d_{\text{TV}}(\mu_1, \mu_2) \leq \mathbb{P}(X_1 \neq X_2)}$$

for every coupling (X_1, X_2) with distributions μ_1, μ_2.

There also exists an optimal coupling for which this inequality becomes an equality.

Intuition. The quantity d_{TV} is defined on the event space $(S, \mathcal{S})$ and examines the greatest difference between X_1, X_2 there. By contrast, $\mathbb{P}(X_1 \neq X_2)$ is the probability of an event in the space Ω: that X_1, X_2 take different values. In brief, d_{TV} is the difference between the probabilities of events on S, the output space, whereas $\mathbb{P}(X_1 \neq X_2)$ is the difference in the input to the random experiment, on Ω, with $X : \Omega \to S$.

Proof. If (X_1, X_2) is a coupling of μ_1, μ_2, then for every measurable set $A \subset S$,

$$\begin{aligned} \mathbb{P}(X_1 \in A) - \mathbb{P}(X_2 \in A) &= \mathbb{E}\mathbf{1}_A(X_1) - \mathbb{E}\mathbf{1}_A(X_2) \\ &= \mathbb{E}(\mathbf{1}_A(X_1) - \mathbf{1}_A(X_2)). \end{aligned}$$

We know that whenever $X_1 = X_2$, we have $\mathbf{1}_A(X_1) - \mathbf{1}_A(X_2) = 0$. The converse is not necessarily true.

If we define

$$D = \{\omega \in \Omega : X_1(\omega) \neq X_2(\omega)\},$$

then

$$|\mathbf{1}_A(X_1) - \mathbf{1}_A(X_2)| \leq \mathbf{1}_D.$$

This is a pointwise inequality. If $\omega \notin D$, then $X_1(\omega) = X_2(\omega)$, and we have $0 \leq 0$. If $\omega \in D$, then $X_1(\omega) \neq X_2(\omega)$, and we have $0, 1 \leq 1$.

Therefore,

$$\begin{aligned} |\mu_1(A) - \mu_2(A)| &= |\mathbb{P}(X_1 \in A) - \mathbb{P}(X_2 \in A)| \\ &\leq \mathbb{E}|\mathbf{1}_A(X_1) - \mathbf{1}_A(X_2)| \\ &\leq \mathbb{E}\mathbf{1}_D = \mathbb{P}(X_1 \neq X_2). \end{aligned}$$

Since the inequality above holds for every $A \in \mathcal{S}$, we conclude that

$$d_{\mathrm{TV}}(\mu_1, \mu_2) = \sup_{A \in \mathcal{S}} |\mu_1(A) - \mu_2(A)| \leq \mathbb{P}(X_1 \neq X_2).$$

We do not prove the existence of an optimal coupling because of the complexity of the proof. But we know that such an optimal coupling exists.

The importance of this proposition is that by choosing couplings intelligently, better bounds can be found for the distance between two distributions.

A coupling example

Let $X_1 \sim \mathrm{Ber}(p)$ and $X_2 \sim \mathrm{Ber}(q)$, with $p < q$.

For the independent coupling, that is, under independence,

$$\begin{aligned} d_{\mathrm{TV}}(\mathrm{Ber}(p), \mathrm{Ber}(q)) &\leq \mathbb{P}(X_1 \neq X_2) \\ &= \mathbb{P}(X_1 = 1, X_2 = 0) + \mathbb{P}(X_1 = 0, X_2 = 1) \\ &= (1-p)q + (1-q)p \\ &= p + q - pq. \end{aligned}$$

For example, if $p = 0.5$, $q = 0.6$,

$$d_{\mathrm{TV}}(\mathrm{Ber}(p), \mathrm{Ber}(q)) \leq 0.6.$$

Now define $\theta_1 \sim \mathrm{Ber}(r_1)$ and $\theta_2 \sim \mathrm{Ber}(r_2)$, with $\theta_1 \perp \theta_2$, and set

$$X_1 := \theta_1, \qquad X_2 := \theta_1 \vee \theta_2 \quad \text{(logical OR)}.$$

Since $p < q$, this is intuitively the best way to reduce the difference: whenever $X_1 = 1$, X_2 should also equal one.

We want (X_1, X_2) to be a coupling of $\mathrm{Ber}(p)$ and $\mathrm{Ber}(q)$. Hence,

$$p = \mathbb{P}(X_1 = 1) = \mathbb{P}(\theta_1 = 1) = r_1 \quad \longrightarrow \quad r_1 = p,$$

$$\begin{aligned} q = \mathbb{P}(X_2 = 1) &= \mathbb{P}(\theta_1 = 1) + \mathbb{P}(\theta_1 = 0, \theta_2 = 1) \\ &= r_1 + (1 - r_1)r_2 \quad \longrightarrow \quad r_2 = \frac{q-p}{1-p}. \end{aligned}$$

By the definitions of X_1, X_2,

$$\mathbb{P}(X_1 \neq X_2) = \mathbb{P}(\theta_1 = 0, \theta_2 = 1) = (1 - r_1)r_2 = q - p.$$

For example, $p = 0.5$, $q = 0.6$ gives 0.1.

By calculating d_{TV}, we can see that this is the optimal coupling.

Section 12.4: Wasserstein distances

The **Wasserstein distance** of order $p \in [1, \infty)$ between probability measures μ, ν on $\mathbb{R}$ is defined by

$$W_p(\mu, \nu) = \inf_{(X,Y)\in\Gamma(\mu,\nu)} \big(\mathbb{E}|X - Y|^p\big)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of the probability measures μ, ν.

On closer inspection, we see that Wasserstein distance and total variation distance (TVD) in fact arise from the same idea. The general form of both distances is

$$\inf_{(X,Y)\in\Gamma(\mu,\nu)} \mathbb{E}\big(\text{cost}(X, Y)\big).$$

That is, the infimum of the average transportation cost over all couplings. Their difference is in the form of their cost functions.

If we take the cost function to be $\mathbf{1}_{(X\neq Y)}$, then intuitively the cost of a difference is one, regardless of the magnitude of the difference, and we have

$$\begin{aligned}\inf_{(X,Y)\in\Gamma(\mu,\nu)} \mathbb{E}\big(\text{cost}(X, Y)\big) &= \inf_{(X,Y)\in\Gamma(\mu,\nu)} \mathbb{E}\mathbf{1}_{(X\neq Y)} \\ &= \inf_{(X,Y)\in\Gamma(\mu,\nu)} \mathbb{P}(X \neq Y).\end{aligned}$$

This is the same proposition as in the previous section.

Now, if we take the cost to be $|X-Y|^p$, of order p, we arrive at the definition of Wasserstein distance. The pth root converts the magnitude to a distance; the unit of the function is p-dimensional.

We state the following proposition without proof:

"The Wasserstein distance of order p is a metric on the space of all distributions on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ for which the moments of order p are finite."

When $p = 2$, we therefore take the cost to be d^2, with d the Euclidean distance. This means that the transportation cost between two points, representing distributions, is proportional to the square of their distance. Intuitively, moving more mass over a given distance, or over a greater distance, requires more energy. This gives it a physical interpretation:

$$E = \frac{1}{2}mv^2.$$

Quantile function

The quantile function is the generalized inverse of the cumulative distribution function, CDF. Thus $Q(a)$, for $0 \leq a \leq 1$, gives the point before which a fraction a of the mass lies.

Proposition, without proof: A practical way to calculate Wasserstein distance

For probability measures μ_1, μ_2 on $\mathbb{R}$, the Wasserstein distance of order $p \in [1, \infty)$ is calculated by

$$\boxed{W_p(\mu_1, \mu_2) = \left(\mathbb{E}(Q_1(U) - Q_2(U))^p\right)^{1/p}},$$

where Q_1, Q_2 are the quantile functions of μ_1, μ_2, and U is a uniform random variable on $(0, 1)$.

This proposition states that $(Q_1(U), Q_2(U))$ is an optimal coupling for Wasserstein distance of order p.

We do not prove this proposition, but it is intuitively understandable. The best coupling pairs positions that are equal in terms of mass. For example, the position at 0.2 of the mass of the first distribution is coupled with the position at 0.2 of the mass of the second, and their distance is calculated. Note the quantile function and the uniform distribution on $(0, 1)$. Since the cost function is of the form d^p, with $p \geq 1$, it is understandable that the formula above represents an optimal coupling.

To understand this better, we examine an example.

Consider three cities A, B, C located at $-x, 0, x$, respectively. Also consider the two probability measures

$$\mu = \frac{1}{2}\delta_{-x} + \frac{1}{2}\delta_0 \quad \text{(supply)}, \qquad \nu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_x \quad \text{(demand)}.$$

They can be interpreted as follows:

μ: one half is produced in city A, at position $-x$, and one half in city B, at position 0.

ν: one half is consumed in city B, at position 0, and one half in city C, at position x.

Now consider the couplings γ_0, γ_1 below, on random vectors in $\mathbb{R}^2$:



We have

$$\begin{aligned} \mathbb{P}(X_0 \neq Y_0) &= \frac{1}{2}, \\ \mathbb{P}(X_1 \neq Y_1) &= 1, \end{aligned} \tag{*1}$$

and

$$\begin{aligned} \left(\mathbb{E}|X_0 - Y_0|^2\right)^{1/2} &= \sqrt{2}\, x, \\ \left(\mathbb{E}|X_1 - Y_1|^2\right)^{1/2} &= x. \end{aligned} \tag{*2}$$

It is easy to check that $d_{\mathrm{TV}} = \frac{1}{2}$, so γ_0 is an optimal coupling for total variation distance, by (*1). It can also be checked that γ_1 is the quantile coupling of the two distributions. Thus, by (*2), it is an optimal coupling for W_2.

The purpose of this example is to show that optimal couplings for TVD and Wasserstein distance may be different.